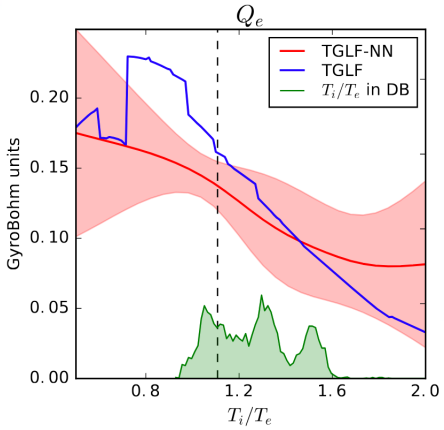
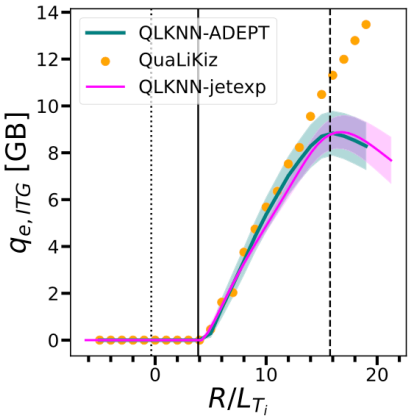
Error bars in NN Surrogate Models for TGLF





TGLF-NN (Meneghini, NF 2017)

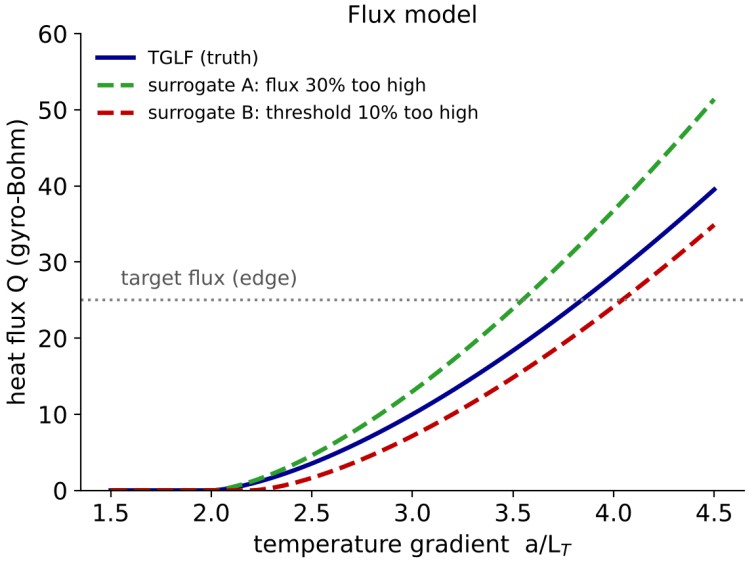
- Sensitivity scan of fluxes
- **Green histogram is distribution of points in the DB**
- **Red area is the ensemble ambiguity of TGLF-NN**
  - Trained 20-30 NNs on same TGLF data
  - Prediction = average of NNs
  - Error bars = std deviation

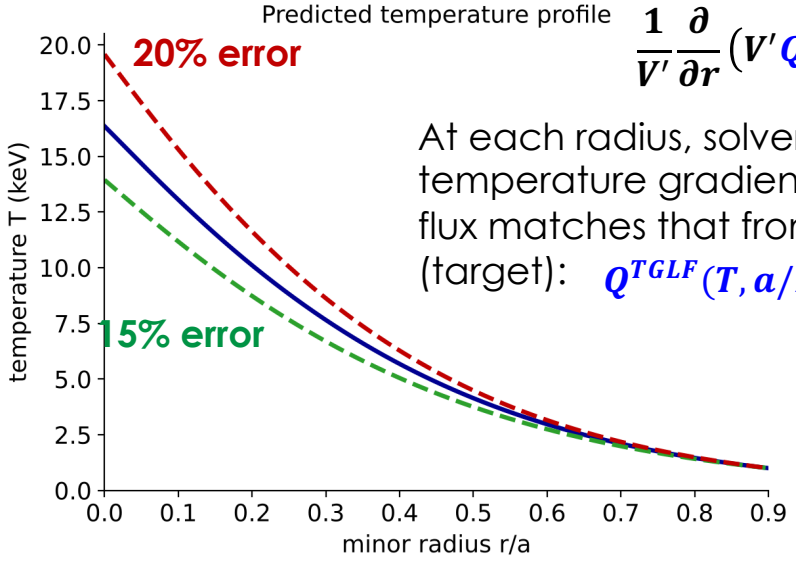
ADEPT for QuaLiKiz (Zanisi, NF 2024)

- Same idea: Ensemble of networks
- Added classifier: stable or turbulent
- Uncertainty grows away from training data and peaks near "critical" gradient
- Active learning: run QuaLiKiz where networks disagree most, retrain, repeat
- 200,000 runs matched surrogate trained on 20 million random runs

Same steps as Task 3: an error bar for each prediction → check it → use it.

How surrogate error changes the predicted profile





$$\frac{1}{V'}\frac{\partial}{\partial r}\left(V' Q^{TGLF}\right) = S$$

At each radius, solver finds temperature gradient where TGLF flux matches that from the source (target): $Q^{TGLF}(T, a/L_T) = Q^{target}$

Flux 30% too high: Curve is steeper, so same target flux is reached at smaller a/LT. Gradient moves ~8%, even though flux was ~30% wrong. "Stiffness" absorbs most of the error.

Threshold 10% too high: Whole curve shifts to the right, so every gradient shifts by same amount. A 10% error moves the solution by that much at every radius.

Gradient errors add up toward the core: Temperature is found by integrating the gradient inward from the edge. So small gradient error at each r builds into a large error at the center (r/a=0).